



Innovation & Entrepreneurship Hub for Educated Rural Youth SURE ProEd (formerly SURE Trust)

Customer and Revenue intelligence

The domain of the Project:

DATA ANALYTICS

GROUP -1

Team Mentors (and their designation):

Ms. Nithyasri Kannathal EL

Senior Product Engineer at EsperGroup

&

Mr. Bhargavesh Dakka

GenAI Consultant at Deloitte USI

Submitted By

Bhingi Manoj

B.Tech in Electronics & Communication Engineering

Ilahia College of Engineering & Technology

Period of the Project

July 2026 to September 2026



*Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd (formerly SURE Trust)***

Declaration

The project titled **Customer and Revenue intelligence** has been mentored by **Ms. Nithyasri Kannathal EL** ma'am and **Mr. Bhargavesh Dakka** sir, organised by SURE Trust, from February 2026 to September 2026 for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Submitted By:

MANOJ- G1 DA

Mentor's Name: Ms. Nithyasri Kannathal EL

Designation - Senior Product Engineer at EsperGroup

Mentor's Name: Mr. Bhargavesh Dakka

Designation - GenAI Consultant at Deloitte

Prof. Radhakumari

Executive Director & Founder

SURE Trust



Innovation & Entrepreneurship Hub for Educated Rural Youth SURE ProEd (formerly SURE Trust)

Executive Summary

1. Introduction

1.1 Background

1.2 Problem Statement

1.3 Project Scope

2. Project Objectives

2.1 Main Objectives

2.2 Expected Outcomes

3. Methodology and Results

3.1 Data Preparation

3.2 Feature Engineering

3.3 Fraud Detection Model

3.4 Risk Scoring and Priority Levels

3.5 Streamlit Application

3.6 Email Alerts and Data Storage

3.7 Investigator Feedback and Retraining

3.8 System Architecture

4. Social and Industry Relevance

4.1 Business Relevance

4.2 Responsible Use

5. Learning and Reflection

5.1 Technical Learning

5.2 Internship Experience

6. Future Scope and Conclusion

Executive Summary



*Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd (formerly SURE Trust)***

The **Automated Customer & Revenue Intelligence Data Pipeline** is an end-to-end data engineering and analytics project developed using the **AdventureWorks dataset**. The project is designed to transform raw customer, product, sales, returns, territory, and calendar data into reliable, business-ready information for decision-making.

The solution is built on **Databricks** using **PySpark**, **Delta Lake**, and a **Bronze–Silver–Gold architecture**. The Bronze layer stores raw data, the Silver layer performs data cleaning and validation, and the Gold layer provides summarized business data. **Incremental data loading** is implemented to process new or updated records efficiently.

A **Star Schema** consisting of fact and dimension tables is created to support analytical reporting. The Gold-layer data is connected to **Power BI** to provide dashboards covering revenue, profit, customers, products, territories, returns, and key business KPIs.

The project also incorporates **Machine Learning** for customer churn prediction using a Random Forest model, with **MLflow** used for experiment tracking. A **Streamlit application** provides an interactive interface for business analysis, while an **AI Assistant** enables users to ask business questions in natural language.

To reduce manual intervention, **Databricks Workflows** are used to automate the execution of the data pipeline through scheduled tasks and task dependencies.

Overall, the project demonstrates a complete real-world data solution integrating **data engineering, data analytics, business intelligence, machine learning, AI, and workflow automation** to convert raw data into actionable customer and revenue insights.

Introduction

1.1 Background



*Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd (formerly SURE Trust)***

In today's data-driven business environment, organizations generate large volumes of data from customers, products, sales transactions, returns, and different business regions. However, raw data alone does not provide meaningful insights unless it is properly collected, cleaned, transformed, integrated, and analyzed.

Traditional data processing approaches often involve manual data preparation and separate reporting systems, which can lead to delays, inconsistent data, and difficulty in identifying important business trends. Businesses therefore require automated and scalable data pipelines that can continuously process data and provide reliable information for decision-making.

This project, **Automated Customer & Revenue Intelligence Data Pipeline**, addresses these challenges by using **Databricks, PySpark, Delta Lake, and a Bronze–Silver–Gold architecture** to process AdventureWorks data. The project incorporates **incremental loading and Databricks Workflows** to reduce manual processing and automate pipeline execution.

The processed data is modeled using a **Star Schema** and presented through **Power BI dashboards**. In addition, **Machine Learning** is used for customer churn prediction, while **Streamlit and an AI Assistant** provide an interactive way to explore business insights.

1.2 Problem Statement



Innovation & Entrepreneurship Hub for Educated Rural Youth SURE ProEd (formerly SURE Trust)

Organizations generate large volumes of customer, product, sales, and returns data, but raw data is often stored in separate files and requires significant manual processing before it can be used for analysis. This can result in **data quality issues, delayed reporting, inefficient data processing, and difficulty identifying customer and revenue trends.**

The absence of an automated and centralized data pipeline also makes it difficult to efficiently process **new or updated records** and maintain consistent business reporting.

Therefore, there is a need for an **automated, scalable, and reliable data pipeline** that can:

- Integrate data from multiple sources.
- Clean, validate, and transform raw data.
- Process new and updated records using incremental loading.
- Organize data into an analytical Star Schema.
- Provide reliable revenue, customer, product, territory, and returns insights.
- Identify customers who are at risk of churn using Machine Learning.
- Automate pipeline execution and reduce manual intervention.
- Provide interactive dashboards and AI-assisted business analysis.

This project addresses these challenges by developing an **Automated Customer & Revenue Intelligence Data Pipeline** using **Databricks, PySpark, Delta Lake, Power BI, Machine Learning, MLflow, Streamlit, AI, and Databricks Workflows.**

1.3 Project Scope

The scope of the **Automated Customer & Revenue Intelligence Data Pipeline** is to develop an end-to-end data platform that converts raw AdventureWorks business data into reliable, actionable customer and revenue insights.

The project covers:

- **Data Ingestion:** Collect customer, product, sales, returns, territory, and calendar data from CSV files.
- **Data Engineering:** Process and transform data using **Databricks and PySpark.**
- **Data Lakehouse:** Implement **Delta Lake** with a **Bronze–Silver–Gold architecture.**
- **Incremental Loading:** Process new and updated records efficiently without reprocessing the entire dataset.
- **Data Quality:** Handle missing values, duplicates, invalid records, and data inconsistencies.
- **Data Modeling:** Build a **Star Schema** with fact and dimension tables for analytical reporting.
- **Business Analytics:** Generate revenue, profit, customer, product, territory, and return-related KPIs.
- **Machine Learning:** Develop a **customer churn prediction model** using historical customer behavior.
- **ML Tracking:** Use **MLflow** to track machine-learning experiments and model performance.
- **Visualization:** Create interactive **Power BI dashboards** for business users.



*Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd (formerly SURE Trust)***

- **Application:** Develop a **Streamlit application** for interactive business analysis.

1.2 Problem Statement

Organizations generate large volumes of **customer, product, sales, and returns data**, but this data is often stored in separate files and requires manual processing before it can be used for analysis. This can lead to **data quality issues, delayed reporting, inefficient processing, and difficulty identifying important customer and revenue trends**.

There is also a challenge in efficiently processing **new and updated records** without repeatedly processing the entire dataset.

Therefore, there is a need for an **automated and scalable data pipeline** that can:

- Integrate data from multiple sources.
- Clean and validate raw business data.
- Process new and updated records using incremental loading.
- Organize data into an analytical Star Schema.
- Generate reliable revenue, customer, product, territory, and returns insights.
- Identify customers at risk of churn using Machine Learning.
- Provide interactive dashboards and business insights.
- Automate the complete data processing workflow.

This project addresses these challenges by developing an **Automated Customer & Revenue Intelligence Data Pipeline** using **Databricks, PySpark, Delta Lake, Power BI, Machine Learning, MLflow, Streamlit, AI, and Databricks Workflows**.

1.3 Project Scope

The **Automated Customer & Revenue Intelligence Data Pipeline** focuses on building an end-to-end data engineering and analytics solution using the AdventureWorks dataset.



*Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd (formerly SURE Trust)***

The project includes:

1. **Data Ingestion** – Ingest customer, product, sales, returns, territory, and calendar data.
2. **Data Processing** – Clean and transform data using Databricks and PySpark.
3. **Lakehouse Architecture** – Implement Bronze, Silver, and Gold layers using Delta Lake.
4. **Incremental Loading** – Process new and updated records efficiently.
5. **Data Modeling** – Build a Star Schema with fact and dimension tables.
6. **Business Analytics** – Generate revenue, profit, customer, product, territory, and returns KPIs.
7. **Machine Learning** – Predict customer churn using historical customer behavior.
8. **Power BI** – Build interactive dashboards for business analysis.
9. **Streamlit Application** – Provide an interactive interface for exploring business insights.
10. **AI Assistant** – Allow users to ask business questions using natural language.
11. **Automation** – Automate pipeline execution using Databricks Workflows.

1.4 Project Limitations

- **Dataset Limitation** – The project uses the AdventureWorks dataset, which may not represent the complexity and volume of real-world enterprise data.
- **Real-Time Limitation** – The current solution uses **batch/incremental processing and scheduled workflows**, not true real-time streaming.
- **Data Quality Limitation** – The quality of insights depends on the accuracy and completeness of the source data.
- **ML Limitation** – The churn model is based on historical customer behavior and may not accurately predict every customer's future behavior.
- **Revenue Limitation** – Revenue calculations may use available product/list-price information rather than actual historical transaction prices.
- **AI Limitation** – AI-generated recommendations depend on the quality of the Gold-layer data and the capabilities of the selected AI model.
- **Deployment Limitation** – The Streamlit application is primarily designed as a project/demo application rather than a fully production-grade enterprise system.
- **Scalability Limitation** – The project demonstrates scalable architecture, but extensive testing with very large enterprise-scale datasets is outside the current scope.

1.5 Innovation Component

The key innovation of the **Automated Customer & Revenue Intelligence Data Pipeline** is the integration of **data engineering, analytics, machine learning, AI, and automation** into a single platform.



*Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd (formerly SURE Trust)***

- **Automated Incremental Processing** – Processes new and updated records without repeatedly processing the complete dataset.
- **Medallion Architecture** – Uses Bronze, Silver, and Gold layers to improve data quality and reliability.
- **Unified Analytics Platform** – Combines Databricks, Star Schema, and Power BI for structured business intelligence.
- **Predictive Customer Intelligence** – Uses Machine Learning to identify customers who are likely to churn.
- **AI-Powered Business Assistant** – Allows users to ask business questions in natural language and receive insights from trusted Gold-layer data.
- **Interactive Analytics Application** – Streamlit provides an additional user-friendly interface beyond traditional dashboards.
- **Automated Workflow** – Databricks Workflows automatically executes the pipeline according to task dependencies and schedules.

Project Objectives

2.1 Main Objective



Innovation & Entrepreneurship Hub for Educated Rural Youth SURE ProEd (formerly SURE Trust)

The main objective of this project is to **develop an automated Customer and Revenue Intelligence platform that transforms raw business data into reliable, actionable insights** using modern data engineering, analytics, machine learning, and AI technologies.

Key Objectives

- Build an automated **Databricks data pipeline** using PySpark.
- Implement **Bronze–Silver–Gold architecture** with Delta Lake.
- Process **new and updated data using incremental loading**.
- Create a **Star Schema** for efficient analytics.
- Analyze **customers, revenue, products, sales, and returns**.
- Predict **customer churn using Machine Learning**.
- Develop **Power BI dashboards** for business decision-making.
- Provide a **Streamlit application and AI Assistant** for interactive insights.
- Automate the complete pipeline using **Databricks Workflows**.

2.2 Specific Objectives

The project aims to develop an automated and intelligent platform that can transform raw business data into meaningful customer and revenue insights. The specific objectives are:

- **Data Integration:** To collect and integrate customer, product, sales, returns, territory, and calendar data from different sources.
- **Data Pipeline Development:** To build an automated data processing pipeline using **Databricks and PySpark**.
- **Data Management:** To implement **Bronze, Silver, and Gold layers** using Delta Lake for reliable and structured data processing.
- **Incremental Processing:** To process new and updated records efficiently using **incremental loading**.
- **Data Quality:** To clean, validate, and transform data by handling missing values, duplicates, and inconsistent records.
- **Data Modeling:** To design a **Star Schema** with fact and dimension tables for efficient analytical reporting.
- **Business Intelligence:** To calculate important KPIs such as revenue, profit, orders, customers, sales quantity, and returns.
- **Predictive Analytics:** To develop a **customer churn prediction model** using Machine Learning and track experiments using MLflow.
- **Visualization:** To create interactive **Power BI dashboards** that help users understand customer and revenue performance.
- **Interactive Application:** To develop a **Streamlit application** for exploring business data and insights.
- **AI Integration:** To provide an **AI Assistant** that can answer business-related questions using trusted data.



*Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd (formerly SURE Trust)***

- **Automation:** To automate pipeline execution and task dependencies using **Databricks Workflows**.

2.3 Expected Outcomes and Deliverables

The project is expected to deliver an **automated customer and revenue intelligence platform** that converts raw business data into reliable and actionable insights.

- **Automated Data Pipeline:** A complete pipeline will be developed using Databricks and PySpark to process business data efficiently.
- **Reliable Data Layers:** Clean and structured Bronze, Silver, and Gold datasets will be available using Delta Lake.
- **Incremental Data Processing:** New and updated records will be processed without unnecessarily reprocessing the complete dataset.
- **Improved Data Quality:** Missing, duplicate, and inconsistent records will be identified and handled during data processing.
- **Analytical Data Model:** A Star Schema will provide structured fact and dimension tables for efficient analysis.
- **Business Insights:** Revenue, profit, sales, customer, product, territory, and returns performance will be available through analytical tables and KPIs.
- **Customer Churn Prediction:** A Machine Learning model will identify customers who are potentially at risk of churn.
- **Interactive Dashboards:** Power BI dashboards will provide clear visual insights for business decision-making.
- **Interactive Web Application:** A Streamlit application will provide users with an easy way to explore business insights.
- **AI-Based Insights:** An AI Assistant will allow users to ask business questions in natural language.
- **Automated Execution:** Databricks Workflows will reduce manual intervention by automatically executing the pipeline according to defined schedules and dependencies.

Methodology and Results

3.1 Project Workflow



Innovation & Entrepreneurship Hub for Educated Rural Youth SURE ProEd (formerly SURE Trust)

The project starts by collecting the AdventureWorks customer, product, sales, returns, territory, and calendar data. The raw files are ingested into **Databricks** using PySpark and stored in the **Bronze layer** using Delta Lake.

The data is then processed through **incremental loading** to efficiently handle new and updated records. In the **Silver layer**, the data is cleaned, validated, transformed, and prepared for analysis. A **Star Schema** is then created using fact and dimension tables, followed by the **Gold layer**, which contains business-ready data and important KPIs such as revenue, profit, customers, products, and returns.

The processed data is used for **Power BI dashboards** and the **Streamlit application** to provide interactive business insights. A **Machine Learning model** is also developed to predict customer churn, with MLflow used for experiment tracking. Finally, **Databricks Workflows** automate the execution of the complete pipeline through scheduled tasks and dependencies.

3.2 Dataset and Input Structure

The project uses the **AdventureWorks dataset**, a sample retail and e-commerce dataset containing information about customers, products, sales, returns, territories, and dates.

The dataset is used to build the complete **Customer & Revenue Intelligence Data Pipeline**. The data is ingested into Databricks, processed through the Bronze–Silver–Gold architecture, and then used for analytics, customer churn prediction, Power BI dashboards, Streamlit, and AI-based insights.

Main Dataset Files

- **Customers** – Customer personal and demographic information.
- **Products** – Product details, categories, prices, and costs.
- **Product Categories** – Product category information.
- **Product Subcategories** – Product subcategory information.
- **Sales 2015, 2016, 2017** – Historical sales transaction data.
- **Returns** – Information about returned products and quantities.
- **Territories** – Geographic and regional sales information.
- **Calendar** – Date information used for time-based analysis.

3.3 Data Preparation and Validation

The AdventureWorks dataset is first prepared by **checking the structure, data types, missing values, duplicates, and invalid records**. The required CSV files are loaded into Databricks using PySpark, and



Innovation & Entrepreneurship Hub for Educated Rural Youth SURE ProEd (formerly SURE Trust)

column names and formats are standardized. Date fields are converted into proper date formats, while text fields are cleaned and unnecessary spaces are removed.

The data is then validated by checking **primary keys, null values, duplicate records, invalid quantities, and inconsistent relationships** between tables. Records that do not meet the required quality conditions are corrected, removed, or handled appropriately. After validation, the cleaned data is stored in the **Silver layer** and used for further transformation and analysis.

Key Validation Points

- Check **column names and data types**.
- Identify and handle **missing values**.
- Remove **duplicate records**.
- Validate **primary and foreign keys**.
- Check for **invalid dates and numerical values**.
- Standardize text and categorical values.
- Verify relationships between related tables.
- Perform final **data quality checks** before moving data to the Gold layer.

3.4 Feature Engineering

Feature engineering is the process of **creating useful new features from existing data** to improve business analysis and Machine Learning performance. In this project, customer sales and transaction data are transformed into meaningful customer behavior features.

Key Features Created

- **Recency** – Number of days since the customer's last purchase.
- **Frequency** – Number of orders placed by each customer.
- **Monetary Value** – Total revenue generated by each customer.
- **Average Order Value** – Average revenue generated per order.
- **Total Quantity** – Total number of products purchased.
- **Purchase Frequency** – Average time between customer purchases.
- **Return Quantity** – Total number of products returned.
- **Customer Lifetime Value indicators** – Revenue and profit generated by a customer over time.
- **Churn Label** – Identifies customers as **Active or Churned** based on their future purchasing behavior.
- These engineered features are used for **customer segmentation, churn prediction, business analysis, and personalized decision-making**.



Innovation & Entrepreneurship Hub for Educated Rural Youth SURE ProEd (formerly SURE Trust)

Key Evaluation Metrics

- **Accuracy:** Measures the overall percentage of correctly predicted customers.
- **Precision:** Measures how many customers predicted as churned actually churned.
- **Recall:** Measures how many actual churned customers were correctly identified.
- **F1-Score:** Provides a balance between precision and recall.
- **ROC-AUC:** Measures the model's ability to distinguish between active and churned customers.

3.5 Databricks Model Serving

Databricks Model Serving is used to deploy a trained Machine Learning model as a **REST API endpoint**, allowing other applications to send data and receive predictions.

In this project, the trained **Customer Churn Prediction model** can be registered and tracked using **MLflow**, and then deployed through Databricks Model Serving.

Workflow

Train Model → MLflow Tracking → Register Model → Model Serving Endpoint → Prediction → Streamlit / Application

Benefits

- Deploys the churn model as an **API**.
- Provides predictions on new customer data.
- Allows **Streamlit** or other applications to access the model.
- Supports scalable model inference.
- Makes the ML model easier to integrate into a production application.

3.6 Batch Prediction Process

Batch processing is a method of processing data by **collecting records over a period of time and processing them together at scheduled intervals**, rather than processing each record immediately.

In this project, **Databricks Workflows** are used to automate batch processing. The pipeline processes the available data at scheduled intervals and performs the required transformations through the Bronze, Silver, Star Schema, and Gold layers.

Workflow

Raw Data → Bronze → Incremental Load → Silver → Star Schema → Gold → Analytics



Innovation & Entrepreneurship Hub for Educated Rural Youth SURE ProEd (formerly SURE Trust)

Key Points

- Processes data in **groups or batches**.
- Can be executed on a **scheduled basis**, such as daily.
- Suitable for large volumes of historical data.
- Reduces the need for continuous processing.
- Databricks Workflows automate the batch pipeline.
- In this project, batch processing supports **daily data updates and business reporting**.

3.13 Streamlit Application

The **Streamlit application** is developed as an interactive web-based interface for presenting the customer and revenue insights generated from the Databricks Gold layer. It provides users with an easy way to explore business performance without directly accessing Databricks.

Key Features

- Displays **revenue, profit, orders, and customer KPIs**.
- Provides **customer, product, territory, and returns analysis**.
- Displays **customer churn predictions** from the Machine Learning model.
- Uses interactive **charts, tables, and filters**.
- Connects directly to the **Databricks Gold layer**.
- Provides an **AI Assistant** for natural-language business questions.
- Helps users obtain business insights through a simple web interface.

Workflow

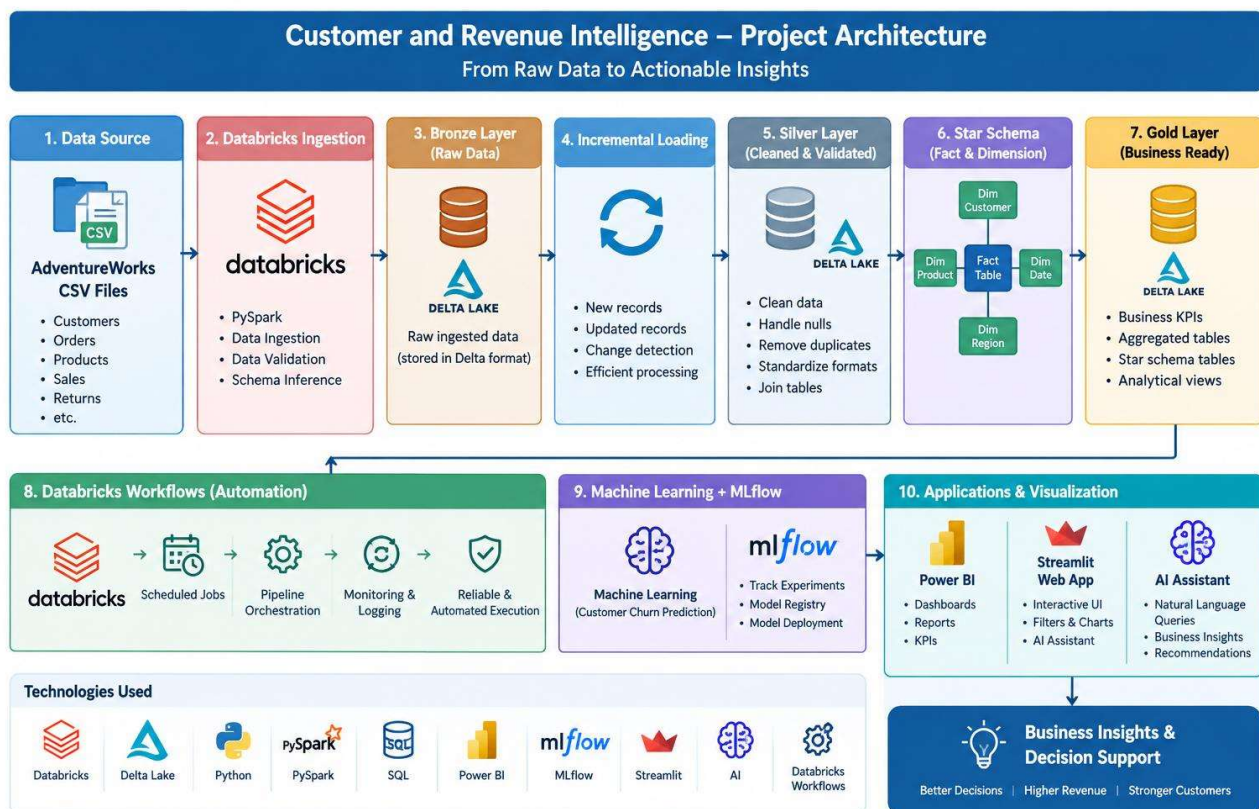
Databricks Gold Layer → Streamlit → Interactive Dashboard → AI Assistant → Business Insights



Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd (formerly SURE Trust)**

3.17 System Architecture

The proposed system consists of multiple layers that work together to transform raw AdventureWorks data into useful business insights. Data is first ingested into **Databricks** and stored in the Bronze layer. Incremental loading is used to process new or updated records. The data is then cleaned and validated in the Silver layer, followed by Star Schema modeling and creation of business-ready Gold tables. The Gold data is used by **Power BI, Streamlit, Machine Learning, and AI** for analysis and decision support. **Databricks Workflows** automate the complete pipeline.





*Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd** (formerly **SURE Trust**)*

Social & Industry Relevance

4.1 Industry Relevance

The project is relevant to the **retail and e-commerce industry**, where organizations manage large volumes of customer, product, sales, and returns data.

- Helps analyze customer purchasing behavior.
- Supports customer churn and retention analysis.
- Helps monitor product and regional performance.
- Supports revenue and profit analysis.
- Enables data-driven business decisions.
- Demonstrates the use of modern data engineering technologies in the industry.

4.2 Business Relevance

The project provides practical value to businesses by converting raw data into **actionable insights and predictive information**.

- Identifies high-value and at-risk customers.
- Helps identify profitable and underperforming products.
- Supports revenue and profit monitoring.
- Helps reduce manual data processing through automation.
- Supports proactive customer retention strategies.
- Provides dashboards and AI-based insights for faster decision-making.

4.3 Decision-Support Value

The project provides decision support by converting raw business data into **clear, actionable insights** that help management make informed decisions.

- Helps identify **high-value and at-risk customers**.



*Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd** (formerly **SURE Trust**)*

- Supports decisions related to **customer retention and churn reduction**.
- Identifies **profitable and underperforming products**.
- Helps analyze **revenue and profit trends**.
- Supports evaluation of **territory and regional performance**.
- Helps monitor **product returns** and identify potential issues.
- Provides **KPIs and dashboards** for faster decision-making.
- Uses **Machine Learning predictions** to support proactive customer management.
- Provides **AI-assisted insights and recommendations** for business users.

4.4 Social and Ethical Relevance

The project has social and practical relevance because it demonstrates how data-driven technologies can help organizations make **better and faster business decisions**. By analyzing customer behavior, sales, revenue, and returns, the system can help businesses understand their customers and improve services.

- Helps organizations make **data-driven decisions**.
- Supports better **customer retention** by identifying potential churn.
- Helps businesses understand **customer needs and purchasing behavior**.
- Identifies **profitable products and high-performing regions**.
- Helps reduce revenue loss through **early identification of business problems**.
- Reduces manual work through **data pipeline automation**.
- Provides accessible insights through **Power BI, Streamlit, and AI**.
- Demonstrates the practical use of modern technologies for **digital business transformation**.

Learning & Reflection

5.1 Technical Learning

During my 6-month internship at **SurePro Organization**, I gained practical technical knowledge in data engineering, analytics, and machine learning.

- Learned **Python, SQL, and PySpark** for data processing and analysis.
- Gained hands-on experience with **Databricks and Delta Lake**.
- Learned **Bronze–Silver–Gold architecture** for data management.
- Implemented **incremental data loading** and data transformation.
- Learned **Star Schema** and analytical data modeling.
- Developed **Power BI dashboards** and business KPIs.
- Gained knowledge of **Machine Learning and MLflow** for customer churn prediction.
- Learned **Streamlit** for developing interactive data applications.



*Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd (formerly SURE Trust)***

- Gained experience with **Databricks Workflows** for pipeline automation.
- Learned how to integrate **AI capabilities** with business analytics.

5.2 Analytical and Problem-Solving Learning

During my internship, I developed analytical and problem-solving skills by working with real-world datasets and identifying data-related issues. I learned to analyze problems systematically, validate data, identify errors, and find appropriate solutions.

- Developed **analytical thinking** through data analysis and SQL queries.
- Improved **problem-solving skills** by resolving data quality and pipeline errors.
- Learned to identify **patterns and trends** from business data.
- Practiced breaking complex problems into **smaller, manageable tasks**.
- Gained experience in **debugging and troubleshooting** Databricks and PySpark workflows.
- Learned to make **data-driven decisions** based on analytical results.
- Improved my ability to **interpret business requirements and convert them into technical solutions**.

5.3 Internship Experience

During my **6-month internship at SurePro Organization**, I gained practical experience in **Data Engineering and Data Analytics**. I worked on real-world data processing tasks and developed an end-to-end **Customer and Revenue Intelligence Data Pipeline**.

I gained hands-on experience in:

- Databricks and PySpark
- Data cleaning and validation
- Delta Lake and Bronze–Silver–Gold architecture
- Incremental data processing
- Star Schema and SQL analytics
- Power BI dashboard development
- Machine Learning and MLflow
- Streamlit application development
- AI-assisted business insights
- Databricks Workflows and automation



*Innovation & Entrepreneurship Hub for Educated Rural Youth **SURE ProEd (formerly SURE Trust)***

Future Scope & Conclusion

6.1 Future Scope The project can be further enhanced by introducing advanced technologies and features to make it more scalable and suitable for real-world applications.

- **Real-Time Streaming:** Implement Kafka and Spark Structured Streaming for real-time data processing.
- **Cloud Deployment:** Deploy the complete solution on Azure, AWS, or Google Cloud.
- **Advanced ML Models:** Use advanced algorithms such as XGBoost, Gradient Boosting, or Deep Learning for better churn prediction.
- **Real-Time Model Serving:** Deploy ML models using Databricks Model Serving for real-time predictions.
- **Advanced AI Assistant:** Enhance the AI assistant to provide automated recommendations and business forecasting.
- **Automated Alerts:** Generate alerts for sudden revenue drops, high churn risk, or poor product performance.
- **Scalability:** Extend the pipeline to handle larger datasets and multiple business domains.
- **Advanced Forecasting:** Implement revenue and sales forecasting to support future business planning.

6.2 Conclusion

The **Automated Customer & Revenue Intelligence Data Pipeline** successfully demonstrates how modern data engineering and analytics technologies can be integrated to transform raw business data into meaningful insights. The project uses **Databricks, PySpark, Delta Lake, Bronze–Silver–Gold architecture, incremental processing, and Star Schema** to create a reliable data pipeline.

The processed data is used for **Power BI dashboards, customer churn prediction, MLflow tracking, Streamlit analytics, and AI-assisted business insights**. Databricks Workflows further automate the pipeline and reduce manual intervention.

Overall, the project provides an integrated solution that combines **Data Engineering, Business Intelligence, Machine Learning, AI, and Automation** to support better customer and revenue-related decision-making.



Innovation & Entrepreneurship Hub for Educated Rural Youth SURE ProEd (formerly SURE Trust)

References

1. **Customer and revenue intelligence GitHub Repository:**
[Bmanoj93/mysureproed-project-repo](https://github.com/Bmanoj93/mysureproed-project-repo)
2. **Streamlit AI Live Application:**
[Customer & Revenue Intelligence Platform](#)